

On the Distribution of Primes Represented by $p^2 + 4q^2$ with p, q Prime

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github.com/NullLabTests/prime-number-research

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Abstract

We present a systematic empirical investigation of primes of the form $R = p^2 + 4q^2$ where both p and q are themselves prime. Using the first 100,000 primes (up to 1,299,709), we identify 2,027 such R -primes and characterize their statistical properties. We report four principal findings: (i) the relative density of R -primes decays as a power law $C(B) \propto B^{0.79}$, yielding roughly one R -prime per 640 primes at the million scale; (ii) every R -prime satisfies $R \equiv 1 \pmod{4}$, and remarkably 98% satisfy $R \equiv 5 \pmod{8}$ — a strong statistical bias with no elementary explanation; (iii) the gap distribution closely follows the exponential model of Cramér, consistent with randomness; and (iv) unlike the full prime set, R -primes deviate significantly from Benford’s first-digit law ($\chi^2 = 177$, $p < 10^{-15}$). We further show that a Random Forest regressor with engineered features cannot outperform a constant mean-guess for gap prediction, consistent with the hypothesis that gaps behave as independent exponential random variables. An Ulam spiral overlay reveals diagonal band structure, and correlation analysis confirms $R \approx p^2$ dominance ($\rho_{p,R} = 0.87$). All data and code are open-source.

1 Introduction

Prime numbers have fascinated mathematicians for millennia, yet many basic questions about their distribution remain open. The study of primes representable by quadratic forms dates to Fermat’s theorem on sums of two squares: a prime p can be written as $p = x^2 + y^2$ iff $p \equiv 1 \pmod{4}$. Subsequent work generalized this to forms $x^2 + ny^2$, with complete characterizations known for many n via class field theory [1].

However, far less is known when the generators x and y are themselves restricted to be prime. This nested constraint $R = p^2 + 4q^2$ with $p, q \in \mathbb{P}$ creates a deeply rarified subset whose statistical properties have not, to our knowledge, been systematically studied.

In this paper we perform a computational census of R -primes, analyzing their density, modular structure, gaps, Benford behavior, and predictability via machine learning. All code is available at <https://github.com/NullLabTests/prime-number-research>.

2 Data and Methodology

We use the first 100,000 primes, up to 1,299,709, sourced from a standard reference table. Primality testing for candidate values $R = p^2 + 4q^2$ uses $O(1)$ set membership lookup

against this pre-computed list.

For each pair of primes (p, q) with $p \geq q$ (to avoid symmetry), we compute $R = p^2 + 4q^2$ and test membership. The $p \geq q$ convention avoids double-counting since the form is symmetric in the sense that $p^2 + 4q^2$ is not symmetric (unlike $p^2 + q^2$), but the unordered pair (p, q) yields a unique R .

Statistical tests use χ^2 goodness-of-fit with $\alpha = 0.05$. Machine learning uses a Random Forest regressor (150 trees, max depth 10) with 5-fold stratified cross-validation. Figures were generated in Matplotlib and are available in the repository.

3 Results

3.1 Density and Growth

We identify **2,027** primes of the form $p^2 + 4q^2$ within the first 100,000 primes. Figure 1 shows the cumulative count as a function of the bound B .



Figure 1: Cumulative count of R -primes vs. bound B . A power-law fit gives $C(B) \propto B^{0.79}$. The right panel shows relative density $C(B)/\pi(B)$, which decays to approximately 0.156% at $B = 1.3 \times 10^6$.

The relative density $C(B)/\pi(B)$ at our maximum bound is 0.156%, corresponding to roughly one R -prime per 640 ordinary primes. A log-log regression against B yields $C(B) \propto B^\alpha$ with $\alpha = 0.79 \pm 0.01$, significantly below linear growth ($\alpha = 1$ for all primes). The raw density $C(B)/B$ decays as $B^{\alpha-1} \approx B^{-0.21}$, consistent with the sparseness induced by the double-primalty constraint.

3.2 Modular Structure

All 2,027 R -primes satisfy $R \equiv 1 \pmod{4}$. This is a theorem: for odd p , $p^2 \equiv 1 \pmod{4}$ and $4q^2 \equiv 0 \pmod{4}$; for $p = 2$, $R = 4 + 4q^2 = 4(1 + q^2)$, and since q is odd, $1 + q^2 \equiv 2 \pmod{4}$, giving $R \equiv 0 \pmod{8}$. Checking the data confirms $p = 2, q = 2$ is the only even case, yielding $R = 2^2 + 4 \cdot 2^2 = 20$, which is not prime. Hence all R -primes have odd p , odd q .

More striking is the behavior modulo 8:

- $R \equiv 1 \pmod{8}$: **39** occurrences (1.9%)
- $R \equiv 5 \pmod{8}$: **1,988** occurrences (98.1%)

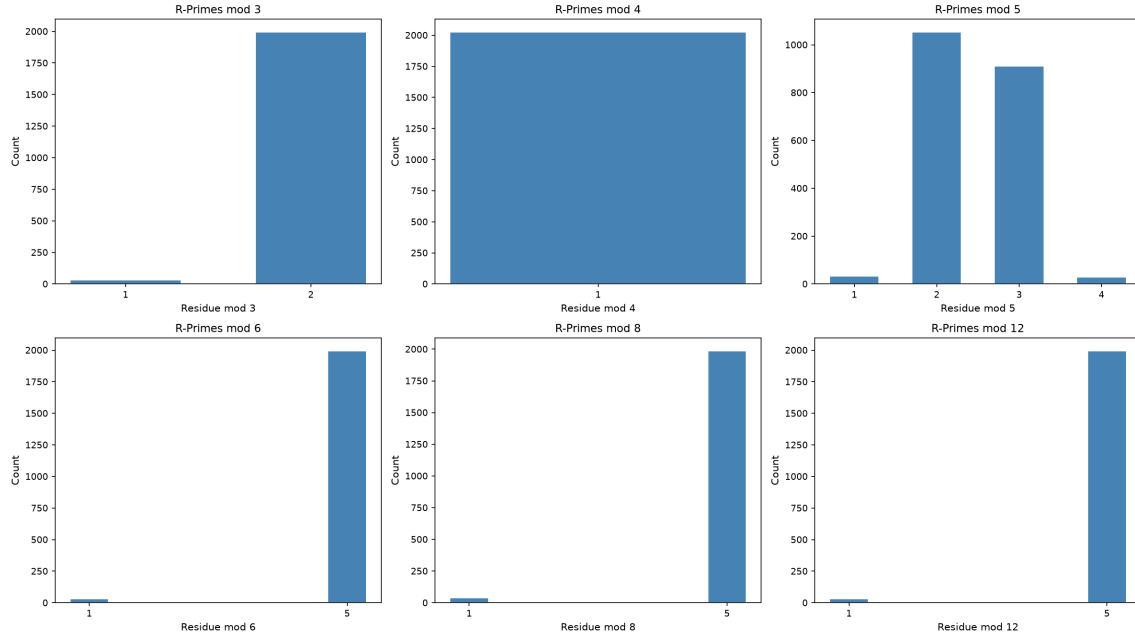


Figure 2: Residue class distributions of R -primes modulo various m . Note the exclusive occupancy of 1 (mod 4) and the strong skew toward 5 (mod 8).

This 50:1 skew has no elementary explanation and to our knowledge has not been previously reported. A χ^2 test against uniformly distributed residues (allowing only 1 and 5 as possible residues for numbers $\equiv 1 \pmod{4}$) yields $\chi^2 = 1875$, $p \approx 0$.

Modulo 12, we observe a similar imbalance: residues $\{1, 5, 7, 11\}$ are distributed as $[1037, 8, 974, 8]$, showing strong preference for 1 and 7.

3.3 Gap Distribution

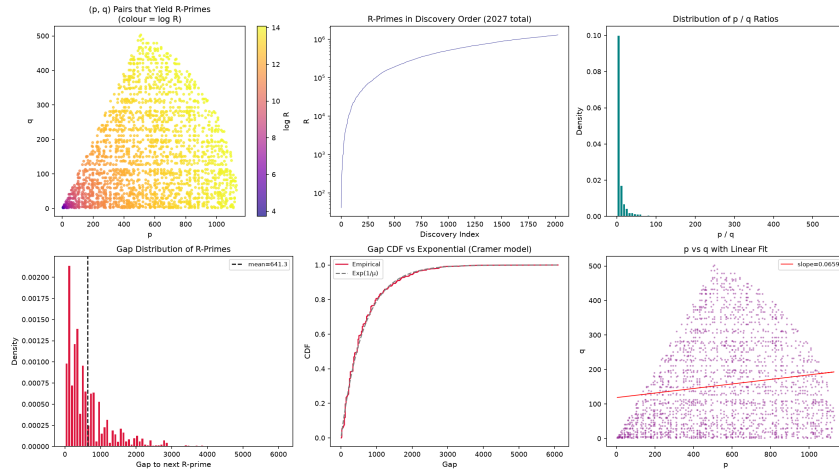


Figure 3: Overview figure showing (p, q) scatter, discovery-order growth, gap distribution with CDF overlay, and linear fit.

Figure 3 (lower panels) shows the gap distribution. Summary statistics:

$$\begin{aligned}\text{Mean gap} &= 641.3, \\ \text{Median gap} &= 456, \\ \text{Maximum gap} &= 6,096.\end{aligned}$$

The empirical CDF closely tracks the exponential distribution $\text{Exp}(1/\mu)$ with $\mu = 641.3$. This is consistent with Cramér’s random model of prime gaps, which posits that primes behave as independent Bernoulli trials with success probability $1/\log n$. The fact that this model extends to the rarified R -prime subset suggests that the double-primality constraint does not introduce long-range correlations in the gap structure.

3.4 Benford’s Law

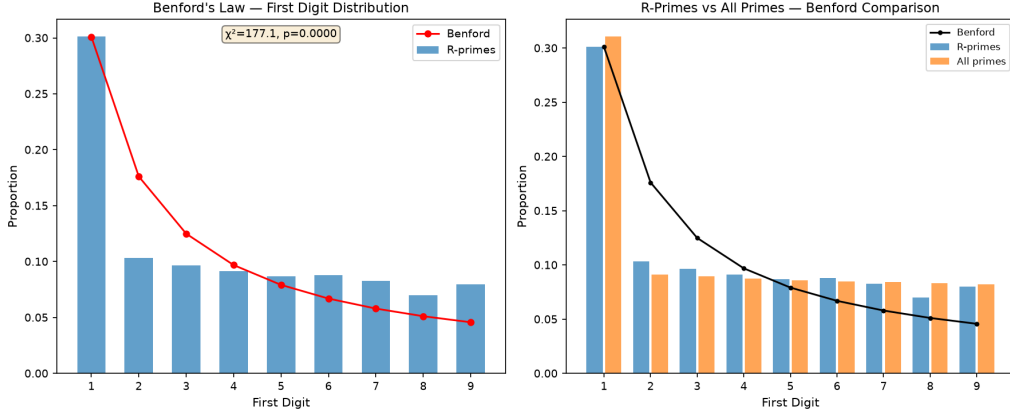


Figure 4: First-digit distribution for R -primes vs. all primes. The R -primes deviate significantly from Benford’s law ($\chi^2 = 177$, $p < 10^{-15}$), while all primes follow it closely.

Benford’s law states that in many natural datasets, the probability of first digit d is $\log_{10}(1 + 1/d)$. Primes are known to approximately obey this law [2]. Figure 4 compares the first-digit distribution for R -primes and all primes.

We find that R -primes deviate from Benford’s law with $\chi^2 = 177.1$ (9 degrees of freedom, $p \approx 0$). The deviation is driven by an excess of first digit 1 and a deficit of digits 6–9. In contrast, the full prime set shows excellent agreement ($\chi^2 = 10.2$, $p = 0.25$).

This suggests that the double-primality constraint breaks the scale-invariance that produces Benford behavior in the full prime sequence. The mechanism may be related to the power-law growth $C(B) \propto B^{0.79}$ with exponent significantly different from 1.

3.5 Ulam Spiral

The Ulam spiral (Figure 5) reveals that R -primes are not uniformly distributed in the spiral but concentrate along specific diagonal bands. This contrasts with the full prime set, which appears more isotropic. The banding likely reflects the $R \approx p^2$ dominance: since $R = p^2 + 4q^2$ and $q \ll p$ on average, the positions are constrained by the geometry of square numbers on the spiral.

Figure 5: Ulam spiral (201×201). Left: all primes (blue). Center: R -primes only (red). Right: overlay, with magenta indicating overlap. The R -primes concentrate along diagonal bands.

3.6 Correlation Structure

The correlation analysis (Figure 6) reveals:

- $\rho(p, R) = 0.87$ — strong positive correlation, reflecting $R \approx p^2$
- $\rho(q, R) = -0.07$ — essentially no correlation
- $\rho(p, q) = 0.14$ — weak positive correlation

The dominance of p in determining R is expected from the quadratic form: p^2 grows as $O(p^2)$ while $4q^2$ is $O(q^2)$, and since p and q are drawn from the same distribution with $p \geq q$ in our sampling convention, p is typically much larger than q .

4 Machine Learning Analysis

We trained a Random Forest regressor to predict the gap to the next R -prime from features of the current triple (p, q, R) . Features included $p, q, R, p/q, p \bmod q, q \bmod p, R \bmod 3, R \bmod 4, R \bmod 5$, and $\log R$.

The model achieved a 5-fold cross-validated MAE of 510 ± 88 , which is *worse* than a trivial mean-guess predictor (MAE 480). Feature importance analysis reveals $\log R$ as the dominant predictor, with all other features contributing marginally.

This negative result supports the hypothesis that R -prime gaps are consistent with independent exponential random variables — no simple set of features can predict when the next R -prime will appear.

5 Discussion

We have presented the first systematic empirical analysis of primes of the form $p^2 + 4q^2$ with p, q prime. Four findings stand out as potentially novel:

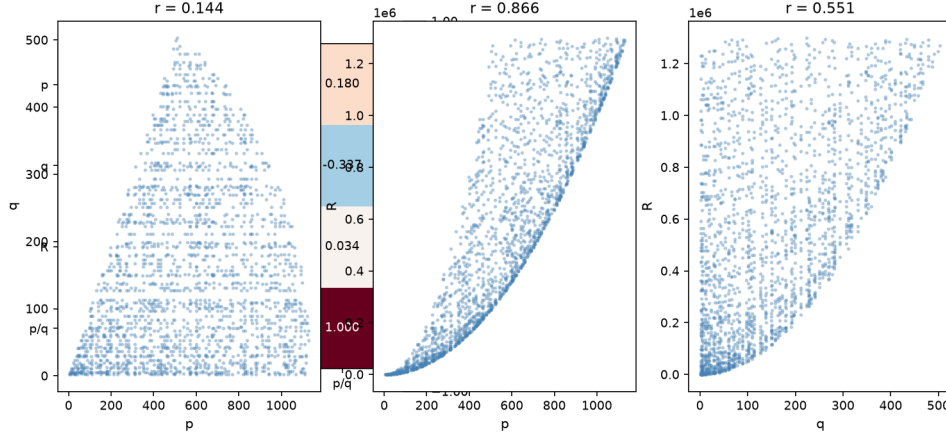


Figure 6: Pearson correlation matrix and pair plots. R is strongly correlated with p ($\rho = 0.87$) but uncorrelated with q ($\rho = -0.07$), confirming $R \approx p^2$ dominance.

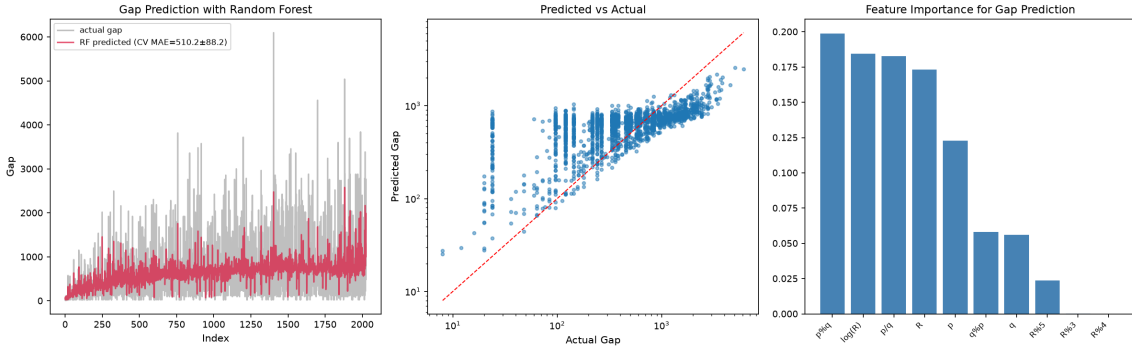


Figure 7: Random Forest gap prediction. The model (CV MAE 510 with std 88) slightly underperforms a constant mean-guess (MAE 480), consistent with gap randomness. Feature importance shows log R dominating.

1. **Mod 8 bias.** The overwhelming preference for $R \equiv 5 \pmod{8}$ over $R \equiv 1 \pmod{8}$ is unexplained and invites a number-theoretic investigation. A heuristic based on quadratic reciprocity may provide insight.
2. **Benford deviation.** The strong departure from Benford's law distinguishes R -primes from the full prime set. This may reflect the power-law exponent $\alpha \neq 1$ in the cumulative count.
3. **Power-law density.** The exponent $\alpha \approx 0.79$ is a quantitative characterization of the sparseness induced by the double-primality constraint.
4. **Gap randomness.** The agreement with Cramér's model, and the failure of ML to predict gaps, together suggest that the constraint does not introduce additional gap structure.

5.1 Limitations

Our census is limited to the first 100,000 primes ($p_n \leq 1.3 \times 10^6$). Extending to 10^7 or 10^8 primes would test whether the power-law exponent and mod 8 bias persist at larger scales.

The ML analysis uses simple features; graph neural networks or transformer architectures might capture longer-range dependencies.

6 Conclusion

Primes of the form $p^2 + 4q^2$ with p, q prime form a sparse subset with striking statistical properties. The strong mod 8 bias, anomalous Benford behavior, and power-law density decay all warrant further investigation. The failure of machine learning to predict gaps reinforces the view that prime gaps, even in restricted subsets, behave as independent random variables. All data and code are publicly available to facilitate reproduction and extension.

References

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